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SUPERSEDING
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FEDERAL SPECIFICATION

FRYERS, DEEP-FAT (ELECTRIC)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers heavy duty, electrically heated, deep-fat fryers.

1.2 Classification. The fryers shall be of the following types, sizes, models, grades, styles and classes specified (see 6.2).

Type I - Counter mounted

Size 1 - Twenty pounds production capacity - fat capacity, not less than 12 pounds 1/

Grade A - Stainless steel exterior
Grade B - Carbon steel, nickel and chrome plated

Type II - Floor model

Size 2 - Thirty pounds production capacity - fat capacity, not less than 15 pounds
Size 3 - Sixty pounds production capacity - fat capacity, not less than 28 pounds
Size 4 - Ninety pounds production capacity - fat capacity, not less than 40 pounds
Size 5 - One hundred twenty-five pounds production capacity - fat capacity, not less than 50 pounds

Model L - Leg mounted
Model C - Caster mounted
Grade A - Stainless steel exterior
Grade B - Carbon steel, gray or black baked-enamel exterior

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Type III - Stand model (modular)

Size 2 - Thirty pounds production capacity - fat capacity, not less than 15 pounds
Size 3 - Sixty pounds production capacity - fat capacity, not less than 28 pounds
Size 4 - Ninety pounds production capacity - fat capacity, not less than 40 pounds
Size 5 - One hundred twenty-five pounds production capacity - fat capacity, not less than 50 pounds

Grade A - Stainless steel exterior

Type IV - Drop-in counter model

Size 1 - Twenty pounds production capacity - fat capacity, not less than 12 pounds
Size 2 - Thirty pounds production capacity - fat capacity, not less than 15 pounds
Size 3 - Sixty pounds production capacity - fat capacity, not less than 28 pounds
Size 4 - Ninety pounds production capacity - fat capacity, not less than 40 pounds

Grade A - Stainless steel exterior
Grade B - Carbon steel exterior

Type V - Wall mounted (modular) (see 6.3)

Size 3 - Sixty pounds production capacity - fat capacity, not less than 28 pounds
Size 4 - Ninety pounds production capacity - fat capacity, not less than 40 pounds

Grade A - Stainless steel exterior

Type VI - Deck mounted - forty-five pounds production capacity - fat capacity, not less than 28 pounds (submarine use)

Grade A - Stainless steel exterior

Style 1 - Fixed heating elements
Style 2 - Swing-up heating elements

Class 1 - Without automatic fry basket lowering and raising mechanism
Class 2 - With automatic fry basket lowering and raising mechanism

1/ Production capacity is defined as the quantity of raw potatoes (prepared for French frying) that can be processed per hour.

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2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

FF-C-77	-	Casters, Rigid and Swivel (Institutional Duty)
HH-I-521	-	Insulation Blankets, Thermal (Mineral Fiber, for Ambient Temperatures)
QQ-S-698	-	Steel, Sheet and Strip, Low-Carbon
PPP-P-600	-	Porcelain Enamel Products and Household Appliances, Electrical and Mechanical-Requirements for Packaging of

Federal Standard:

FED-STD-123	-	Marking for Shipment (Civil Agencies)

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cummulative bi-monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications, standards, and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston; New York; Washington, DC; Philadelphia; Atlanta; Chicago; Kansas City, MO; Fort Worth; Houston; Denver; San Francisco; Los Angeles; and Seattle, WA.

(Federal Government activities may obtain copies of Federal specifications, standards, and commercial item descriptions, and the Index of Federal Specifications, Standards, and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specification:

MIL-R-11337	-	Ranges, Packaging and Packing of

Military Standards:

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage

(Copies of Military specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

American National Standards Institute (ANSI):

C73 - Dimensions of Caps, Plugs, and Receptacles

(Application for copies should be addressed to the American National Standards Institute, 1340 Broadway, New York, NY 10018.)

American Society for Testing and Materials (ASTM):

A 167 - Stainless and Heat-Resisting Chrominum-Nickel Steel Plate, Sheet, and Strip
A 176 - Stainless and Heat-Resistant Chromium Steel Plate, Sheet, and Strip
A 342 - Permeability of Feebly Magnetic Materials, Tests for

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Sanitation Foundation (NSF):

Standard No. 2 - Food Service Equipment
Standard No. 4 - Commercial Cooking and Hot Food Storage Equipment

Listing of Food Service Equipment

(Application for copies should be addressed to the National Sanitation Foundation, NSF Building, Ann Arbor, MI 48106.)

National Electrical Manufacturers Association (NEMA):

No. ICS - Standards Industrial Controls

(Application for copies should be addressed to the National Electrical Manufacturers Association, 155 East 44th Street, New York, NY 10017.)

Underwriters Laboratories Inc. (UL):

UL 197 - Commercial Electric Cooking Appliances
UL 197-S1 - Commercial Electric Cooking Equipment for Marine Use

(Application for copies should be addressed to the Underwriters Laboratories Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 Standard product. The fryers delivered under this specification shall be the manufacturers commercial product, except for any changes necessary to comply with specification requirements.

3.2 Codes and standards. The stands shall comply with the applicable requirements of NSF Standard No. 2 or 4 and the fryers shall conform to the applicable requirements of NSF Standard No. 4 and UL 197.

3.3 Compliance. Prior to commencing production, the contractor shall submit to the contracting officer, or his authorized representative, satisfactory evidence that the stand and fryer except type VI fryers (see 3.9.5) he proposes to furnish under this specification meet the applicable requirements of NSF and UL as follows:

3.3.1 NSF. Acceptable evidence of meeting the requirements of NSF shall be one of the following:

(1) A listing in the current edition of the NSF "Listing of Food Service Equipment" and display of the NSF seal on the finished stand and fryer.

(2) A certification for the stand and fryer issued by NSF under their special one-time contract evaluation/certification service.

(3) A certified test report from a recognized independent testing laboratory acceptable to the medical department of the service for which the stand and the fryers are being procured indicating that the stand and the fryers have been tested and conform to NSF Standard No. 2 and No. 4, respectively.

3.3.2 UL. Acceptable evidence of meeting the applicable requirements of UL 197 shall be the UL listing mark, or a certified test report from a recognized independent testing laboratory, acceptable to the Government, indicating that the fryers offered conform to the applicable requirements of UL 197 and, when applicable, for shipboard use, UL 197-S1 (Commercial Electric Cooking Equipment for Marine Use.)

3.4 Materials and components. Materials and components shall be as specified herein. Materials not definitely specified shall be of the quality normally used by the manufacturer for electric deep-fat fryers, provided the completed item complies with all provisions of this specification. Commercial tolerances are applicable to all material thickness (see 6.5).

3.4.1 Carbon steel sheet. Carbon steel shall be cold rolled sheet, commercial quality, bright finish conforming to QQ-S-698.

3.4.2 Stainless steel plates, sheets, and strips. The stainless steel plates, sheets, and strips shall conform to type 302 or type 304 of ASTM A 167; or any type in the 400 series of ASTM A 176.

3.4.3 Insulation. Insulation shall be NSF accepted and shall be dry rock wool, glass wool or other types of similar material.

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3.5 Design. Type I fryers shall be designed for counter use. Type II fryers shall be a cabinet type designed for floor use. Type II, fryers, when specified (see 6.2) shall be provided with a solid state control system with or without a computer control system (see 3.5.2.2.1 and 3.5.2.2.2). Type III fryers shall be designed for stand mounting, type IV fryers shall be designed for drop-in counter use, type V fryers shall be designed for wall mounting and type VI fryers shall be designed for deck mounting (submarine use). Model C fryers, shall have casters conforming to type I and type II, class B, wheel diameter 5-inches, style c, synthetic or polyurethane composition rubber mounting optional, of FF-C-77. Two casters shall be type I (rigid) and two shall be type II (swivel) with brakes. Legs, when required, shall be as specified in 3.6.6. Class 1 fryers are provided without automatic fry basket lowering and raising mechanism. Class 2 fryers shall be provided with an automatic fry basket lowering and raising mechanism as specified in 3.6.8.3. The cabinet exterior surfaces back and sides, shall have no protrusions i.e., receptable or connections. The sizes 1, 2, 3, 4, and 5 fryers shall French fry a minimum of 20, 30, 60, 90, and 125 pounds of raw potatoes per hour, respectively, and type VI fryer shall French fry a minimum of 45 pounds of raw potatoes per hour when tested as specified in 4.4.4.

3.5.1 Dimensions. Unless otherwise specified (see 6.2), overall exterior dimensions shall be as specified in table I.

TABLE I. Overall exterior dimension

Fryer size	Height 1/ (from floor to top of cooking vessel)	Maximum width	Maximum depth
	Inches	Inches	Inches
1	N/A (counter model)	18	24
2	33-37	21	31
3, 4 and 5	33-37	24	39

1/ Height for type IV fryers not applicable.

3.5.2 Electrical characteristics. Unless otherwise specified (see 6.2), fryers shall operate on a nominal 208-volt, 3 phase, 60-Hertz (hz) system. All fryers except type II, model C fryers shall have provisions for permanent electrical connection. Provisions for grounding shall be included for each fryer. All type II, model C, size 2 and size 3 fryers designed for 208 volt, 3 phase, 60 hz, 4 wire operation shall be provided with a UL approved power supply cord not less than 5-1/2 feet nor more than 6 feet long and NEMA 15-60 plug.

3.5.2.1 Contactors. One or more contactors shall carry the resistive load of the fryer. If the contactor is intended for mounting outside the fryer it shall conform to applicable NEMA standards. Contactors shall be magnetic type, rated up to 250 volts per pole. The rated voltage of the contactor coil shall be the same as the applied control circuit voltage.

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3.5.2.2 Control system.

3.5.2.2.1 Solid-state. The solid-state control system shall incorporate as a minimum, a temperature dial, power ON-OFF switch, start/reset switch, test switch, melt/fry switch and fuse holder and fuse. All switches shall be located on a single control panel located on the front to the fryer. Switches to be of the toggle, rocker or membrane type. Toggle type switches when provided in fryers for naval shipboard installations shall be protected with a gasket. As each system is energized, the indicator light for that function shall operate as intended by the manufacturer, i.e., glows, flashes or goes out.

3.5.2.2.2 Solid-state computer. The solid-state computer control system shall incorporate the solid-state control system specified in 3.5.2.2.1 with the addition of a computer system. The computer solid-state system shall incorporate two independent computer systems with a single or dual computer control with one computer system controlling the lift basket and the second computer system controlling the right basket. Each computer system shall consist of a computer "ON" switch, computer "ON" indicating light and an audible alarm. The audible alarm shall sound whenever the food product is done and the computer indicating light functions. A control panel located on the fryer front shall contain all the switches and controls and shall be removable with the use of simple tools.

3.6 Construction.

3.6.1 Vessels.

3.6.1.1 Cooking. Cooking vessels shall be fabricated of carbon steel as specified in 3.4.1 or stainless steel as specified in 3.4.2. For Navy procurement a stainless steel cooking vessel shall be provided. The carbon steel cooking vessel may be nickel plated at the manufacturer's option. Prior to nickel plating, copper plating shall not be allowed. The cooking vessel for the type I, size 1, type II, sizes 2 and 3, and type III, sizes 2 and 3, type IV, sizes 1, 2, and 3 and type V, size 3 and type VI fryers shall be removable or nonremovable type. Type II, sizes 4 and 5, and type III, sizes 4 and 5, type IV, size 4 and type V, size 4 fryers shall have a nonremovable cooking vessel. Removable cooking vessels shall be provided with handles for carrying and pouring the fat or shall be provided with means to drain the hot fat. The upper portion of the vessels shall be formed to provide a free board height to prevent overflow of frying component when tested as specified in 4.4.4.2. Drainage shall be arranged from the foaming area back to the vessel. Space shall be provided in the bottom for accumulation of crumbs. Unless otherwise specified (see 6.2), type IV, class 2 fryers shall have two half size removable cooking vessels in lieu of one large vessel. Each vessel shall operate independently of the other. Cooking vessels for the type VI fryers shall be designed so that when in use with the normal amount of fat the fat will not overflow when the fryer is at an angle of 30 degrees. The cooking vessel shall measure 10 inches wide by 14 inches deep by 14 inches high with a tolerance of plus or minus 1/8 inch. The rear wall of the cooking vessel shall be provided with a die stamped oil capacity indicating mark. The indicating mark shall include a dotted or horizontal line and the words "OIL LEVEL or FAT LEVEL".

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3.6.1.1.1 Material thicknesses. Stainless steel vessels for the type II, sizes 4 and 5, and the type III, sizes 4 and 5, type IV, size 4 and type V, size 4 fryers shall have a thickness of 0.0562 inch. All other stainless steel vessels shall have a thickness of 0.043 inch except when the cooking vessel is provided with stiffening ledges and flanges, the thickness shall be 0.037 inch. Carbon steel vessels for the type II, sizes 4 and 5 fryers shall have a thickness of 0.059 inch. All other carbon steel vessels shall have a thickness of 0.0478 inch except when the cooking vessel is provided with stiffening ledges and flanges, the thickness shall be 0.0359 inch.

3.6.1.2 Fat-draining vessels. When drain facilities are provided (see 3.6.1.3) a fat draining vessel shall be furnished. The fat draining vessel including the handles shall be fabricated of stainless steel. The fat draining vessel shall be provided with handles for carrying and pouring the hot fat. The capacity of the fat draining vessel shall be not less than 50 percent of the normal operating cooking vessel capacity for the type I fryers and not less than 35 percent of the normal operating cooking vessel capacity for all other type fryers when tested in accordance with 4.4.5.

3.6.1.3 Drain. Facilities shall be provided to drain the cooking vessel of fryers having a nonremovable vessel and also on removable vessels when handles for carrying and pouring the fat are not furnished. The drain may consist of a hand operated valve of not less than 1-inch nominal pipe size. When draining is accomplished with a hand valve, the means for opening and closing the valve shall not interfere with the removal of the fat-draining vessel.

3.6.2 Baskets. Unless otherwise specified (see 6.2), each class 1 fryer shall be equipped with one large and two small fry baskets. The class 2 fryer shall be equipped with two small fry baskets. Baskets for type VI fryers shall be provided with folding handles. Fry baskets shall be fabricated of carbon-steel conforming to 3.4.1, or stainless steel conforming to 3.4.2 wire, or expanded-metal. Opening in the wire baskets shall be approximately 1/4 inch square. Diamond-shaped openings of the expanded-metal baskets shall have a short dimension between 3/16 and 5/16 inch. Carbon-steel baskets, after fabrication, shall be tin plated, and then retinned by the hot-dip method or may be nickel plated except no copper plating shall be used prior to nickel plating. The large basket shall be slightly smaller than the frying area. The two small baskets shall be suitable for simultaneous use in the fryer. Each basket shall be 5-1/2 to 6-1/2 inches deep with supporting frame and one or more handles. Basket supports or other suitable device shall be furnished to permit draining the grease from the cooked food in the basket, or baskets, into the cooking vessel without spilling grease over the sides.

3.6.3 Cover. Each fryer shall be equipped with a removable cover with a recessed handle for covering the cooking vessel. Grade A fryers shall be provided with a stainless steel cover conforming to 3.4.2 and grade B fryers shall be provided with a stainless steel cover conforming to 3.4.2 or nickel-plated carbon-steel cover. No copper plating shall be used under nickel plating.

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3.6.4 Cabinet. Grade A fryers shall be encased with stainless steel conforming to 3.4.2 having a minimum thickness of 0.0375 (20 Gauge) inch. Grade B fryers shall be encased with carbon steel sheets having a minimum thickness of 0.0359 (20 Gauge) inch. The cabinet shall enclose all working parts. Cabinets shall be of a frame or frameless construction. When frameless construction is used, the panel sides shall be formed to serve as frames or otherwise reinforced as necessary to comply with paragraph 4.4.6. When a frame construction is used, the materials shall be fabricated as specified in paragraphs 3.4.1 or 3.4.2. For type II fryers, that portion of the cabinet facing the operator shall be provided with a shelf or may be provided with other means at the bottom to accommodate the fat-draining vessel. The shelf shall be fabricated from the same material and thickness as the cabinet. The cabinet front shall be provided with one or more doors or when specified (see 6.2), the cabinet shall be provided with an open front. The shelf shall be constructed to support the fat-draining vessel when filled with the specified amount of fat (see 3.6.1.2), and shall not become distorted or show any deflection. The cabinet bottom shall be provided with means of reinforcement for leg installation to form a rigid assembly. When legs are attached directly to the body they shall be attached to a structural member or suitable reinforcing member or reinforcing plate. Attaching holes in the cabinet bottom may be tapped. Threaded fasteners, i.e., captured nuts, reinforcing plate with a threaded form a nut, bolt or stud, or by other means welded or secured to the cabinet bottom may be utilized for leg attachment. For cabinets of fryers for Navy shipboard use (see 3.9). The cabinet including legs or casters shall not become distorted or show any deflection when tested in accordance with 4.4.6. The cabinet bottom for the type II fryers shall be reinforced at the four corners for leg or caster mounting. Each corner shall be reinforced by any one of the following: (a) A combination of an adapter plate 0.119 inch thick and triangle gusset 3/16 inch thick with 1/4-20 threaded hole; (b) Extruded plate 0.119 inch thick with a threaded form nut (1/4-20); (c) Channel (U-shaped) with a bolt (1/2-13) welded to the channel; (d) Combination of a plate, gusset with a threaded fastener welded or bolted to the bottom of the cabinet bottom. The edge of the cabinet bottom may be provided with a return for reinforcement.

3.6.5 Stand (modular fryer). The stand for the type III fryers shall be of a welded or bolted construction and shall consist of a 90 degree angle, rectangular frame reinforced with corner braces; or bell shaped under table leg gussets, threaded flange, welded or bolted to the frame and detachable tubular legs, fabricated from stainless steel specified in 3.4.2. The legs shall conform to the requirements of 3.6.6. The frame shall be constructed from 90 degree angle, with the exterior surfaces forming the top, sides, and ends of the stand. The reinforcing corner braces and leg gussets shall be welded or bolted to the four inside corners of the retangular frame. When of a welded construction, the gussets and all seams, open joints or edges shall be welded with corrosion-resisting weld material. When of a bolted construction, lock nuts or other secure type of lock fasteners shall be used. All exterior welded areas shall be ground flush. The completed stand, with legs attached, shall be a minimum height of 18-1/2 inches and the stand shall not protrude at any point beyond the fryer body (i.e. length or width). The stand shall not become distorted or show any deflection when tested in accordance with 4.4.6.

3.6.6 Legs. The body of the type II model L and the type III fryers shall be supported on four adjustable, stainless steel detachable tubular type legs secured to the bottom of the rectangular frame. The leg tube thickness shall be 0.065 (16 Gauge) inch wall. The legs for the type II fryers shall be not less than 6 inches

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long and the legs for the type III fryers shall be not less than 16 inches long. Legs shall be equipped with leveling devices, adjustable with a range not less than 1 inch. For leg mounting of fryers for Navy shipboard installation (see 3.9).

3.6.6.1 Wall bracket assembly. Type V wall mounted fryers shall be provided with a wall bracket and in-wall carrier assembly. The bracket shall be fabricated from not less than 0.140 inch thick stainless steel or carbon steel with a stainless steel exterior facing. Carrier shall be fabricated from 5 inch by 1-7/8 inch by 3/8 inch carbon steel channels having two horizontal in-floor members not less than 24 inches long and two vertical in-wall members of suitable length to obtain the fryer height specified in 3.5.1. Carrier vertical framing shall be braced and welded at top and bottom, and shall be provided with bolts or studs located and sized to accommodate the wall bracket. The wall bracket and in-wall carrier assembly shall not become distorted or show any deflection when tested in accordance with 4.4.6.

3.6.7 Heating elements. Heating elements for all fryers shall be of the enclosed, resistance type having nickel-plated steel or stainless steel sheaths. Copper plating shall not be allowed. Heating elements, when in operating position, shall be located either across the bottom or parallel to the walls of the cooking vessel or a combination of both of these. When the heating elements are in more than one horizontal plane, there shall be a minimum space of 1/2 inch between the lower element and the bottom of the cooking vessel and minimum space of 1/2 inch between elements. A cool zone shall be provided below the elements into which loose food particles may settle and accumulate without being burned or recirculated. When elements are located at the bottom, a nickel plated or stainless steel removable or movable protector shall be provided. Heating elements may be inserted through the front or at the top of the vessel, individually or as a complete unit, and shall permit complete maintenance from the front or top. Style 2 fryers shall have a device to hold or lock the heating elements in the raised position. When heating elements on style 2 fryers are designed not to operate out of the oil, an automatic cutoff shall be provided to disconnect power to the elements. Style 2 fryers shall have the swing-up element hinged area attached to a structural member.

3.6.8 Thermostat. An adjustable thermostat shall be provided with a minimum range from 200° to 375°F. The thermostat shall have a positive "OFF" position and be marked at 50 degree intervals. A separate "ON-OFF" switch may be provided in addition to or in lieu of the positive "OFF" position to control power to the thermostat. Thermo-stat and switch shall be located so as to be readily accessible to the operator. The thermostat shall maintain an average temperature of 350° $\pm$ 5°F when tested as specified in 4.4.2.1. The thermostat shall limit the maximum temperature to 425°F. The temperature amplitude of the thermostat shall not exceed 25°F, when tested as specified in 4.4.2.2. Instructions for calibrating the thermostat shall be provided with each fryer (see 3.8.2).

3.6.8.1 Overtemperature thermostat. The deep fat fryers shall be provided with a second or overtemperature thermostat. The first thermostat shall be as specified in 3.6.8. The second or overtemperature thermostat shall be nonadjustable, type. The overtemperature circuit shall be manual reset type installed in the temperature control system to shut off the entire electrical system when the fat reaches a temperature of 430°F thru 460°F and shall be tested in accordance with 4.4.3. In lieu thereof, a fusible thermostatic cutoff is acceptable where temperature controls

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are electric. All heating circuit control devices shall be enclosed within the body of the devices or in a separate enclosure forming part of the body of the device. For Naval shipboard installation (see 6.2), in addition to the adjustable operating thermostat, the fryer shall be equipped with an upper temperature limit thermostat to serve as a temperature limit switch. This protective thermostat shall not interfere with the normal cycling of the fryer under control of the operating thermostat, but shall operate in the event of above normal fat temperature, closing when the fat reaches a temperature not exceeding 430°F. This thermostat shall be a nonadjustable thermostat with normally open contacts. The thermostat shall be of the "negative bias" type, so constructed that, in event of hydraulic failure of the thermostat, the contacts will close and activate the shunt trip coil of an external disconnect circuit breaker. The contacts of this thermostat shall be connected to a separate terminal block within the fryer housing. (This secondary safety thermostat is intended for connection to the 120 volt shunt trip coil of an external disconnect circuit breaker, when fryer is installed). When the electric circuitry of solid state control fryers is specifically approved by the Naval Sea Systems Command for self contained, integral oven temperature limit controls, the separate terminal block for connection to an external shunt trip circuit breaker is not required. Such controls must still provide for connection to the control circuit of an external fire protection system.

3.6.8.1.1 Primary thermostat shunt by pass assembly. The deep fat fryer shall be provided with a primary thermostat shunt by pass assembly. The shunt by pass assembly shall consist of a normally open, momentary contact, oil tight, push button type switch with a flush head, signal light and wiring. Whenever the shunt by pass switch is depressed the primary thermostat shunt by pass circuit shall be energized and the signal light shall glow and deenergize whenever the shunt by pass switch is released or the overtemperature thermostat is operated when tested as specified in 4.4.3.

3.6.8.2 Signal lights. The fryer shall be equipped with a signal light to indicate that power is available at the fryer when "ON-OFF" switch is in the "ON" position. The fryer shall have a signal light that glows when the heating elements are heating and goes out whenever the dialed temperature is reached. Signal lights shall be of such size and located in such a manner as to be readily visible by the operator.

3.6.8.3 Automatic fry basket mechanisms. Class 1 fryers shall be provided without an automatic fry basket lowering and raising mechanism. Class 2 fryers shall be provided with automatic fry basket lowering and raising mechanisms capable of automatically lowering the fry basket into the cooking vessel and raising the fry basket at completion of the cooking cycle. Class 2, sizes 1 and 2 fryers shall be provided with one automatic fry basket lowering and raising mechanism and sizes 3, 4, and 5 fryers shall have two automatic lowering and raising mechanisms each capable of independently automatically lowering the fry basket into the cooking vessel and raising the fry basket at completion of the cooking cycle. Class 2 fryers shall be provided with two small fry baskets. The fry baskets when not in the cooking mode shall be in the raised position. Each automatic fry basket mechanism shall consist of an adjustable timer, and an activating cooking cycle switch. Each adjustable timer shall have a time range from 1 to 15 minutes for selecting the cooking time.

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Each adjustable timer shall have an integral push button cycling switch, that electrically activates the automatic fry basket mechanism. It shall not be necessary to reset the adjustable timer after each cooking cycle. When the preset time cycle is complete, the fry basket mechanism shall automatically raise and remain in the raised position until the push button is activated again. Fryer controls, switches, moving parts, and electrical circuits shall operate through a complete cooking cycle without malfunction, binding or excessive looseness when tested as specified in 4.4.7. (A complete cooking cycle of the fry basket is defined as the lowering of the fry basket into cooking vessel and remaining there until the time cycle is complete then elevating the fry basket to the raised position.)

3.6.8.4 Contract data requirements. The contract data requirements shall be as specified (see 6.2 and 6.2.1).

3.6.9 Performance.

3.6.9.1 Heating rate. The fryers shall heat the rated charge of fat from 75° to 375°F in not more than 15 minutes with an ambient temperature of 75° ± 5°F, when tested as specified in 4.4.1. In addition to attaining this temperature within the specified time, a quantity of fat shall be melted to permit submergence of the fry baskets for cooking.

3.6.9.2 Kilowatt input rate. The fryer shall be so constructed that the power required to maintain a temperature of 375° ± 5°F, for 1 hour at an ambient temperature of 75° ± 5°F, with the fryer idling under control of the thermostat and with the cover removed, shall not exceed the kilowatt hourly input rates specified in table III when tested as specified in 4.4.1. The maximum input kilowatt rate shall be as specified in table III.

TABLE III. Kilowatt rates

Fryer	Kw. hour input	Kw. input rating (maximum)
Size 1	0.600	6.0
Size 2	1.000	12.5
Size 3	1.000	12.5
Size 4	1.500	22.0
Size 5	1.500	22.0
Type VI	0.680	12.5

3.6.9.3 Frying capacity. The fryers shall maintain an average frying compound temperature of 300°F or above, and shall be capable of recovery to 375°F prior to completion of frying cycle when tested as specified in 4.4.4. The capacity of the types II, III, and IV, size 2 fryer shall not be less than 45 pounds per hour.

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3.7 Finish. Grade A fryers shall be finished in accordance with the requirements of NSF No. 4 on the exterior surfaces. Type I, grade B fryers shall have a nickel and chrome plated finish in accordance with the manufacturer's standard practice. Types II and IV, grade B fryers, shall be finished in either gray, dark green or black enamel in accordance with the manufacturer's standard practice. Fryers shall be smooth on all surfaces and edges and shall be free from discoloration or stains.

3.8 Marking. Identification and operating instructions shall be permanently and legibly marked directly on the item or on an aluminum, brass, or stainless steel plate firmly affixed to the item. Marking shall be stamped, embossed, engraved or by photosensitive means. The plate thickness shall be not less than 0.012 inch.

3.8.1 Identification. Each fryer identification plate shall include the manufacturer's model and seriel number, name, trade name or trademark, of such known character as to be readily identifiable to the manufacturer. In addition, such information as required by UL 197 for the fryer shall be included on the plate.

3.8.2 Users instruction plate. All fryers except type IV fryers shall have the instruction plate located on the outside front. Type IV fryers shall have the instruction plate provided for installation on face of counter. The words "DEEP FAT FRYER" shall be not less than 3/16 inch high and the words "USER'S INSTRUCTIONS", shall be not less than 1/8 inch high. All other characters shall be not less than 3/32 inch high. The plate shall be of sufficient overall size to contain the following information etched, embossed, or die stamped thereon in legible characters:

DEEP FAT FRYER

USER'S INSTRUCTIONS

1. If provided with drain valve, close valve tightly before filling cooking vessel to proper level with oil or fats.

2. Turn on heating element, set thermostat for fat temperature desired, and heat for about 15 minutes before starting to cook food. Do not turn on the heating elements unless they are covered with fats or oils.

3. After food is fried, drain basket by hanging it on basket support.

4. Turn off heating element immediately when through using fryer.

5. Do not allow burned crumbs to accumulate.

6. Do not let grease accumulate and harden on frame and body of cooking vessel.

7. Remove fat and all residue from cooking vessel after each day's use, using scraper or soft brush if necessary, then wash with hot soapy or detergent water, rinse with clear hot water and dry thoroughly. Do not return fat to vessel prior to next time fat is used.

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8. Before fat is returned to cooking vessel, it shall be filtered.

9. Clean cooking vessel at least once each week. Fill vessel with solution of mild washing compound, bring to boil, and let stand for several hours. Refill with clear water and boil again. Add 2 ounces of vinegar to the final cold-water rinse to neutralize any trace of alkalinity remaining from compound. Stir, drain, rinse, and wipe vessel with soft cloth.

10. Clean exterior of fryer with a detergent solution of grease solvent.

11. Cover cooking vessel when not is use.

3.8.3 Instructions for calibration of thermostat. Explicit instructions for calibration and adjustment of the thermostat shall be furnished with each fryer.

3.9 Navy shipboard installation. For Navy shipboard installation (see 6.2), dimensional requirements for fryers shall be as specified in figures 1, 2, and 3 and guard rails, leg mounting, electrical characteristics, incline operations and environmental suitability shall be as follows:

3.9.1 Guard rails. The type II and type VI fryers shall be provided with a stainless steel guard rail extending across the full width at the fryer front. The guard rail shall extend 3 to 4 inches beyond the front of the fryer and in the same plane as the fryer top. The guard rail shall be supported by brackets.

3.9.2 Legs and mountings. The bottom of the legs for the type I, size 1, type II and type VI fryers shall be provided with holes for mountings. The type I, fryer leg mounting holes shall be as specified in figure 1. The type II and type VI fryers shall have leg mounting holes as specified in figures 3 and 2 respectively. All legs shall be fabricated of stainless steel and the thickness shall not be less than 0.078 inch. Legs shall be reinforced without cross bracing to form a rigid assembly.

3.9.3 Electrical requirements. Type I, type II and type VI fryers shall operate on a nominal 440-volt, balanced 3-phase, 60 Hertz (hz) system. The fryers shall be equipped with a 440/110 volt step down transformer to allow a single 440-volt, 3 phase, 60 Hz connection for the 110-volt control circuit. Provisions for grounding shall be included for each fryer. Access to controls and connections shall be gained from the front of the fryers. Mercury type of controls and switches shall not be used. Electrical components such as cutouts with fuses, contactors, etc. shall be mounted in an enclosed compartment within the fryer body. Exposed terminals in the drainage area within the body of the fryer enclosure shall be concealed by a cover. Electrical connections shall be of a screw, bolt, or plug-type that will not loosen.

3.9.4 Inclined operation. The type I, type II, type IV and type VI fryers shall operate satisfactorily when inclined at an angle of 15 degrees each side of the vertical in each of two vertical planes at right angles to each other, with no spillage of fluid or product when tested as specified in 4.4.8.

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3.9.5 Environmental suitability. Fryers shall be capable of withstanding normal shipboard vibration and mechanical impact. Fryers that have passed Navy Hi-shock tests (see 6.4) or are approved by the Naval Ship Engineering Center Electric Equipment Branch or its field representative as being similar to fryers that have Hi-shock tests are considered to be sufficiently substantial in construction to withstand shipboard environmental conditions without further tests. The fryer controls, switches, moving parts, and electrical circuits shall operate through 2 complete cycles without malfunction, binding, excessive looseness, or damage when tested as specified in 4.4.9.

3.9.6 Magnetic permeability. When specified (see 6.2), the complete fryer shall be constructed entirely of 300 series stainless steel specified in 3.4.2. A non-magnetic material is defined as one having a magnetic permeability of not more than 2.0 after fabrication when tested as specified in 4.4.10.

3.9.7 Fryer cover. When specified (see 6.2), the fryer cover shall be designed to provide a smooth flush surface over the entire top of the fryer when not in use. The fryer with the cover in place shall be suitable as a counter top working space. If the design of the cover is such that it cannot be easily grasped for removal, a handle shall be provided.

3.9.8 Thermal insulation. Thermal insulation shall be as specified in 3.4.3 and shall be acceptable by the Naval Ship Engineering Center. The insulation shall have a thermal conductivity not greater than 0.28 British Thermal Unit (BTU) per hour per square foot per 1°F per inch for a mean temperature of 90°F when packed to the density in such a manner that it will not settle or leave portions of the fryer wall uninsulated.

3.9.9 Cabinet enclosed. Fryers shall be provided with cabinet as specified in 3.6.4 with the addition that the cabinet shall be totally enclosed on the back and sides. The cabinet front shall be provided with a door. The cabinet shall have no cracks or crevices which might encourage insect or vermin infestation.

3.9.10 Maintenance. All components requiring repairs, replacement, calibration or any maintenance shall be accessible from the front of the fryer without distributing the fryer mounting devices

3.10 Wiring diagram. The supplier shall furnish, a wiring diagram showing the necessary connections of all electrical components and their service connections. The diagram shall also include all electrical characteristics and wire sizes and types.

3.11 Repair and maintenance. Provisions shall be made to allow ready adjustment, service, and replacement of electrical power system components and operational electrical accessories and components, without disturbing or disassembly of any major component or equipment, as required, for proper operation of the fryer.

3.12 Accessories. When specified (see 6.2), an indicating thermometer of the type specified shall be provided.

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3.13 Workmanship. All components and assemblies of the fryer shall be free from dirt and other extraneous material, burrs, slivers, rough die, tool and grind marks, dents and cracks. Castings, molded parts, and stampings, if used, shall be free of sand, fins, pits, blow holes and sprues. External surfaces shall be free from sharp edges and corners, except when corners are required.

3.13.1 Metal fabrication. Metal used in the fabrication of the fryer shall be free from kinks. Forming and shearing shall not cause damage to the metal.

3.13.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease and other foreign matter. Welds shall be smooth and free of cracks, burn holes, undercuts or incompleted fusion. All scale and flux shall be removed from the finish weld area.

3.13.3 Fastening devices. Threaded fastener and rivet holes shall be accurately punched or drilled and shall have burrs removed. Threaded fasteners shall not be broken, cracked or stripped and shall be drawn tight. Rivets shall fill the hole completely and the heads shall be in full contact with the surface of the member and concentric with the hole.

3.13.4 Wiring. Wiring shall be enclosed, supported, and protected from grease or food spillage and against mechanical damage. Metal edges with which wire comes in contact shall be rounded and smooth.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as other specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.2.2 End item inspection. A lot shall consist of all fryers of one type, size, model, grade, style, and class offered for inspection at one time. The sample unit shall be one completely assembled fryer.

4.2.2.1 Visual examination. The completely assembled fryers shall be examined for defects in table IV. The inspection level shall be II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

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TABLE IV. Classification of visual defects

Examine	Defect	Classification Major	Minor
Finish	Edges and surfaces not smooth	X	
	Not free of discoloration or stains		X
Design, construction, and workmanship (general)	Part or component missing	X	
	Any rough or sharp edges, burrs, slivers, or scratches; rough die or grind marks	X	
	Any component fractured, buckled, bent, punctured, malformed, or damaged	X	
	Component not readily accessible for servicing where required	X	
	Functioning component that does not operate as intended	X	
	Protrusion on cabinet exterior surface	X	
Cooking vessel	Oil level indicating markings missing	X	
Fat draining vessles	Handles not provided	X	
Cover	Recessed handle not provided	X	
Shelf	Any evidence of distortion or deflection when fat draining vessel is filled	X	
Signal light or lights	Not provided (power "ON" or primary thermostat by pass assembly)	X	
Welding	Welds broken, porous, cracked, incomplete, and not ground smooth and polished	X	
	Scale or flux deposits not removed	X	
Controls	Not accessible from front	X	
Primary thermostat	Not provided with primary thermostat by pass assembly	X	
Grounding	Fryers do not have provisions for grounding	X	
Stand (modular fryer)	Not as specified	X	
Legs (as applicable)	Do not provide a rigid support without cross bracing	X	
	Distorted or deflected	X	
	Do not provide means for mounting (shipboard use)	X	

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TABLE IV. Classification of visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Legs (as applicable) (cont'd)	Not adjustable	X	
	Not provided with reinforcement means for leg installation	X	
Guard rails (shipboard use)	Missing	X	
	Loose		X
Hand valve for nonremovable vessel (as applicable)	Interferes with removal of fat draining vessel	X	
Casters	Not provided with brakes as required	X	
Bolts, nuts, screws, studs, and other types of threaded fasteners	Not locking type nuts or fasteners	X	
	Missing, broken, stripped, fractured, or loose	X	
Identification markings	Missing, incomplete, or not legible	X	
Markings (instruction and identification, as applicable)	Missing, incomplete, or not legible	X	
	Not specified type or size, not affixed in prescribed manner		X

4.2.2.2 Dimensional examination. The fryer shall be examined for compliance with dimensions specified. Any dimensions not within the specified tolerance shall constitute a defect. The inspection level shall be S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

4.2.2.3 End item testing. One fryer selected at random from the initial production lot shall be tested as specified in 4.4.1 thru 4.4.5 and 4.4.8 thru 4.4.10. Failure to pass any test shall be cause for rejection of the lot. The contractor shall perform the tests specified in 4.4.6 and 4.4.7 using inspection level S-4 and an AQL of 4.0 defects, expressed in terms of defects per hundred units.

4.2.3 Inspection of preparation for delivery. Preservation, packaging, packing and marking shall be inspected in accordance with the quality assurance provisions in MIL-R-11337.

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4.3 Certificate examination. Certificate of compliance, certified test reports, approved labels or listing marks for codes and standards, as applicable, that are submitted as proof of compliance with the specification requirements shall be examined and validated.

4.4 Methods of inspection.

4.4.1 Heating rate and kilowatt hourly input rate test. The fryer shall be filled with the normal quantity of hydrogenated vegetable oil at an ambient temperature of 75°F ± 5°F. The fryer shall then be connected to the power supply called for in the contract or purchase order. The thermostat shall be set at 375°F and the time required for the temperature to reach 375°F shall be recorded to determine compliance with the heating rate requirement of 3.6.9.1. The maximum input kilowatt rate shall be determined for compliance with 3.6.9.2. The temperature is to be measured in the center of the cooking vessel, 1-inch below the surface of the liquid. After the fryer has reached 375°F, and with the fryer idling under the control of the thermostat and with cover removed, the number of kilowatts hourly input rate specified in 3.6.9.2 shall be determined. Any noncomplince with the requirements specified shall constitute failure of this test.

4.4.2 Thermostat accuracy. The fryer shall be operated with a thermostat dial setting of 375°F when filled with the normal quantity of hydrogenated vegetable oil. The average oil temperature and temperature amplitude shall be determined as specified below. The oil temperature shall be measured at the center of the cooking vessel 1 inch below the oil surface. Inability to meet the thermostat accuracy requirements specified shall constitute failure of this test.

4.4.2.1 Average hydrogenated vegetable oil temperature. The fryer shall be operated until steady oil condition of 375°F is reached. When this condition is reached, the maximum and minimum oil temperatures shall be measured for three consecutive cycles and the average oil temperature determined for compliance with the average oil temperature requirement of 3.6.8. Any noncompliance with the requirements specified shall constitute failure of this test.

4.4.2.2 Temperature amplitude. The maximum and minimum hydrogenated vegetable oil temperatures shall be measured in three consecutive cycles at three control dial settings in the normal usage range of 325° to 375°F after steady oil condition has been reached. The average difference shall be determined, which is the temperature amplitude, to determine compliance with the temperature amplitude requirement of 3.6.8. Inability to meet the temperature amplitude requirements specified shall constitute failure of this test.

4.4.3 Overtemperature thermostat test. The overtemperature thermostatic test shall be performed after completion of the tests specified in 4.4.2.1 and 4.4.2.2 for the first thermostat specified in 3.6.8. The first thermostat shall be rendered inoperative from the automatic control circuit by depressing the primary thermostat shunt by pass switch. With the shunt by pass switch depressed the temperature of the hydrogenated vegetable oil shall be increased until the overtemperature thermostat specified in 3.6.8.1 opens the heating element circuit and the primary thermostat shunt by pass signal light goes out. The temperature of the oil shall be

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measured at the center of the well 1 inch below the surface at 30-second intervals until the temperature stabilizes. The temperature reading of the oil obtained during the observation period shall not exceed the temperature requirement of 3.6.8.1. This test shall be repeated three times for compliance with the by pass requirements of 3.6.8.1.1. Any noncompliance with the requirements specified sall constitute failure of this test.

4.4.4 Frying-capacity test. The frying-capacity test shall be conducted to determine conformance with 3.5 and 3.6.9.3. Any noncompliance with the requirements specified shall constitute failure of this test.

4.4.4.1 Test preparation.

(a) Fresh, white potatoes, commercial grade, for French frying shall be used for this test. Any variety may be used.

(b) Not more than 4 hours before frying, the potatoes shall be peeled, and sliced into 1/2- by 1/2-inch strips. Eyeing and finishing shall not be required. No slivers or small ends shall be included.

(c) Immediately after slicing, the potato strip shall be immersed in 70°F water for not less than 1 hour, and not more than 3 hours.

(d) The soaked potatoes shall be removed from the 70°F water and allowed to drain for 6 minutes at normal room conditions. The fry basket shall be filled with the following weights of drained potatoes:

Fryer	Lbs
Size 1	2.0
Size 2	3.5
Size 3	6.0
Size 4	9.0
Size 5	12.5
Type VI	4.5

(e) The cooking vessel shall be filled on the day of the test with hydrogenated vegetable oil which has not been previously used in the amount indicated in 1.2 for each size fryer.

(f) A single thermocouple shall be positioned 1 inch below the oil surface, halfway between the outer perimeter of the fry basket (when inserted in normal frying position) and the inner surface of the fat container at the rear of the fryer but in no case shall the thermocouple be located closer than 1/2 inch from any heating element.

4.4.4.2 Test method.

(a) With the thermostat set at 375°F, the hydrogenated vegetable oil in the cooking vessel shall be brought up to temperature, allowing the thermostat to recycle for four complete cycles at an oil temperature of 375°F. The fry basket with rated load of potatoes shall be placed in the normal frying position. The thermostat shall be reset at 400°F.

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(b) The oil shall be observed for spillage over the sides of the fryer at any time during the entire frying period to determine compliance with the applicable requirements of 3.6.1.1.

(c) The fry basket shall be removed after exactly 6 minutes in the fryer.

(d) Four consecutive loads of potatoes shall be fried in rapid succession. No temperature readings shall be taken for the first two baskets. Reading shall be the average of the third and fourth baskets (see 3.6.9.3). During the frying period, the oil temperature (as indicated by the average reading of the third and fourth baskets) shall be checked to determine compliance with 3.6.9.3).

(e) After the last load has been removed, the type II, size 5 fryer with an oil temperature of 350°F for 30 minutes shall be checked to determine compliance with the applicable requirements of 3.5.2.1 and 3.5.2.2.2.

4.4.5 Volumetric test for vessels. All nonremovable cooking vessels, (with drain and fat draining vessels), shall be filled with oil at a temperature of 375°F, and the volumetric capacity observed to determine compliance with 3.6.1.1 and 3.6.1.2. Inability to meet the volumetric requirements specified shall constitute failure of this test.

4.4.6 Load test. The cabinets or stands including the legs for the types II, III and VI fryers shall be subjected to a uniform loading equivalent to 125 percent of the total weight. The total weight shall consist of the weight of the fried, hydrogenated vegetable oil and full load of potatoes. When under load the cabinet or stand shall be examined to determine compliance with 3.6.4 and 3.6.5. With the base of the type II fryer in a fixed position subject the fryer to a load of 50 pounds plus or minus 5 percent applied without impact to an area six square inches plus or minus 10 percent for 5 minutes at the top center of the fryer body side. After removing the load, the fryer body side shall be examined for evidence of distortion. This procedure shall be repeated for the opposite side and front of the fryer. In each position distortion shall not exceed 1/2 inch. Wall bracket and in wall carrier assembly for type V wall mounted fryers shall be capable of supporting at least 400 pounds in excess of fryer weight to determine compliance with 3.6.6.1. Inability to meet the load requirements specified shall constitute failure of this test.

4.4.7 Operational test. With the class 2 fryer connected to a rated power source, operate each automatic fry basket mechanism through two complete cooking cycles, for each of the following adjustable timer settings. Timer settings shall be 1, 5, 10, and 15 minutes. Operation shall be checked to determine compliance with 3.6.8.3. Any noncompliance with the performance requirements of 3.6.8.3 shall constitute failure of this test.

4.4.8 Inclined operational test. After the test in 4.4.4.2 is completed, position the fryer with the base set at an angle of 15 degrees, then operate fryer for 30 seconds each at each side of the vertical in each of two vertical planes at right angles to each other. At each of these postions observe for conformance with specified requirements of 3.9.4. Any nonconformance with specified requirements shall constitute failure of this test.

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4.4.9 Environmental test. Before packaging, the fryer shall be subjected to the following vibration and impact tests in accordance with PPP-P-600.

(a) Vibration test. The equipment to be tested shall be placed on the table of the vibration tester. Fences for the equipment being tested may be attached to the test table. Vibration frequency shall be determined by advancing the cycle of vibration until the equipment under test leaves the table sufficiently to allow a thin piece of cardboard, approximately 1/16 inch thick, to be inserted between the bottom edge of the equipment and the platform of the machine. The equipment shall be vibrated for 2 hours.

(b) Impact area. An incline impact testing device of the Conbur type, or other testing machine producing equivalent results, and a shock recorder shall be used. The equipment to be tested shall be fastened on the dolly in such a position that the equipment will make no physical contact with the barrier of the impact machine. The dolly shall be drawn up the incline to a position which will produce a shock upon impact of at least ten times the force of gravity. The test shall be repeated so that the equipment receives two impacts in each horizontal plane.

Failure to operate satisfactorily at the conclusion of the tests or evidence of breakage or disassembly shall constitute failure of this test.

4.4.10 Permeability test. Magnetic permeability shall be tested in accordance with ASTM A 342 to determine compliance with 3.9.6. Any noncompliance with specified requirements shall constitute failure of this test.

5. PREPARATION FOR DELIVERY

5.1 Levels A, B, and C. Fryers shall be preserved, packaged, and packed for each level of protection in accordance with the applicable requirements of MIL-R-11337 (see 6.2). In addition, units having demountable casters shall have the casters removed.

5.2 Marking. Marking shall be in accordance with 5.2.1 or 5.2.2, as specified (see 6.2).

5.2.1 Civil requirements. In addition to any special marking required by the contract, interior packages and shipping containers shall be marked in accordance with FED-STD-123.

5.2.2 Military agencies. In addition to any special marking required by the contract, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The fryers are intended for use in kitchens and galleys for deep-fat frying of various foods. Fryers with leg mountings are intended for use by the Navy for shipboard installation.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

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(a) Title, number, and date of this specification.
(b) Type, size, model, style, grade and class of fryers required (see 1.2).
(c) When for type II, size 5 fryer a solid state control system with or without a computer control system is required (see 3.5).
(d) When other fryer dimensions are required (see 3.5.1).
(e) When other electrical characteristics are required (3.5.2) and when 50 hz operation is required (see 3.5.2).
(f) When the cooking vessel for type IV, class 2 fryers is other than specified (see 3.6.1.1).
(g) When the number and size of baskets, is other than specified (see 3.6.2).
(h) Whether an open front is required for the type II fryer cabinet (see 3.6.4).
(i) When fryer is intended for Naval shipboard installation (see 3.6.8.1 and 3.9).
(j) When fryer is to be constructed of stainless steel for Naval use (see 3.9.6).
(k) When fryer cover is required for Naval use (see 3.9.7).
(l) Contract data requirements (see 3.6.8.4 and 6.2.1).
(m) When the type of indicating thermometer is required (see 3.12).
(n) Selection of applicable levels of preservation, packaging, and packing (see 5.1).
(o) Marking required (see 5.2).

6.2.1 Contract data requirements. Any data items to be delivered under any contract for items covered by this specification should be specifically called for in the contract in accordance with the applicable regulations of the procuring activity (see 3.6.8.4 and 6.2).

6.3 Type V fryers. Purchase of type V fryers, wall mounted with wall bracket assembly, is applicable to new construction or where there is a major structural modification to the wall area which will permit in-wall and in-floor installation of the wall carrier assembly.

6.4 Fryers, Hi-shock. Fryers that have passed the Hi-shock test that was formerly required for fryers under MIL-C-001218B(Ships) or fryers that are satisfactory to the Naval Ship Engineering Center as being similar in construction to fryers that have passed the Hi-shock tests are considered to meet the requirements of 3.9.5 without being tested as described in 4.4.9.

6.5 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.4).

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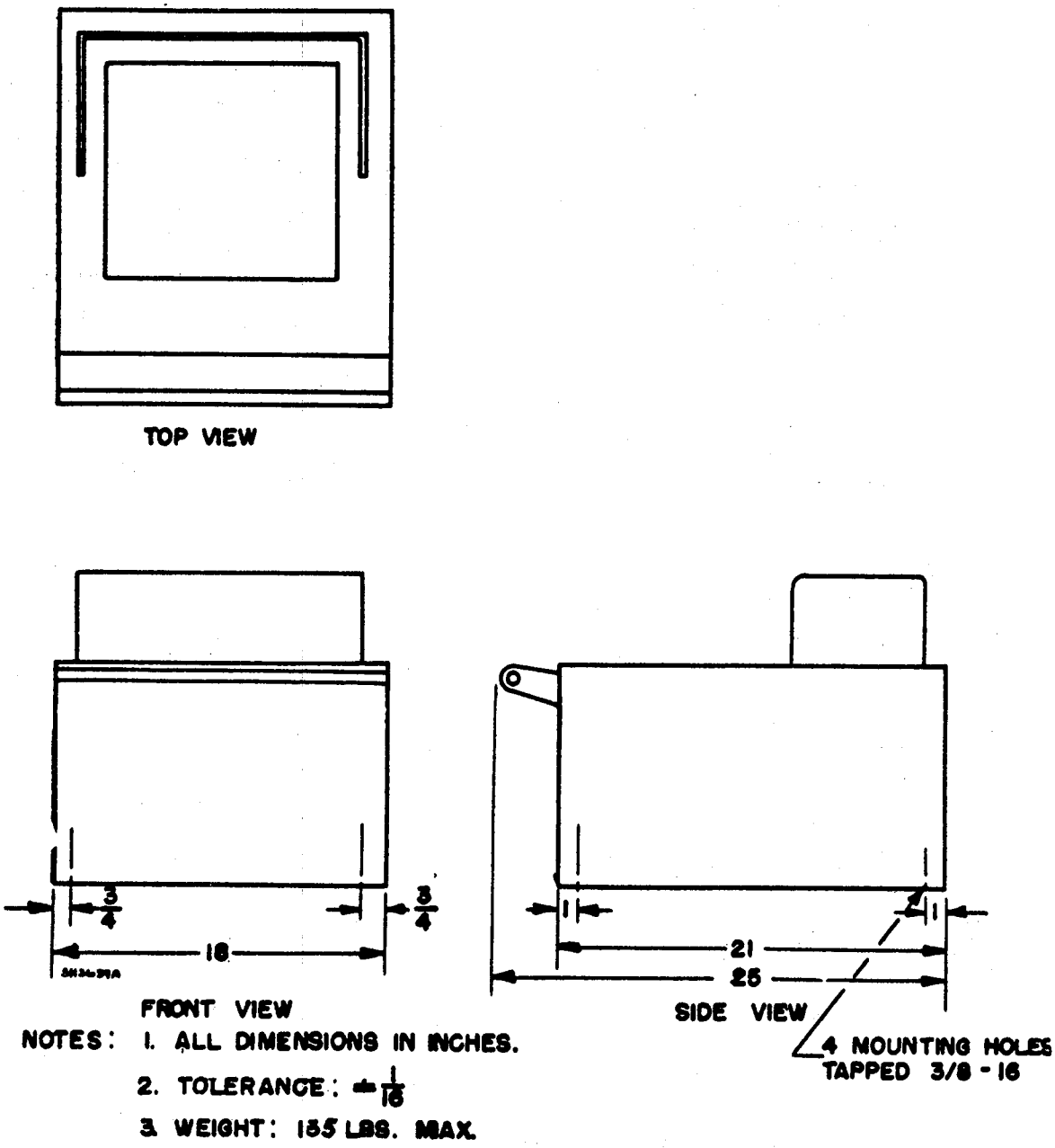
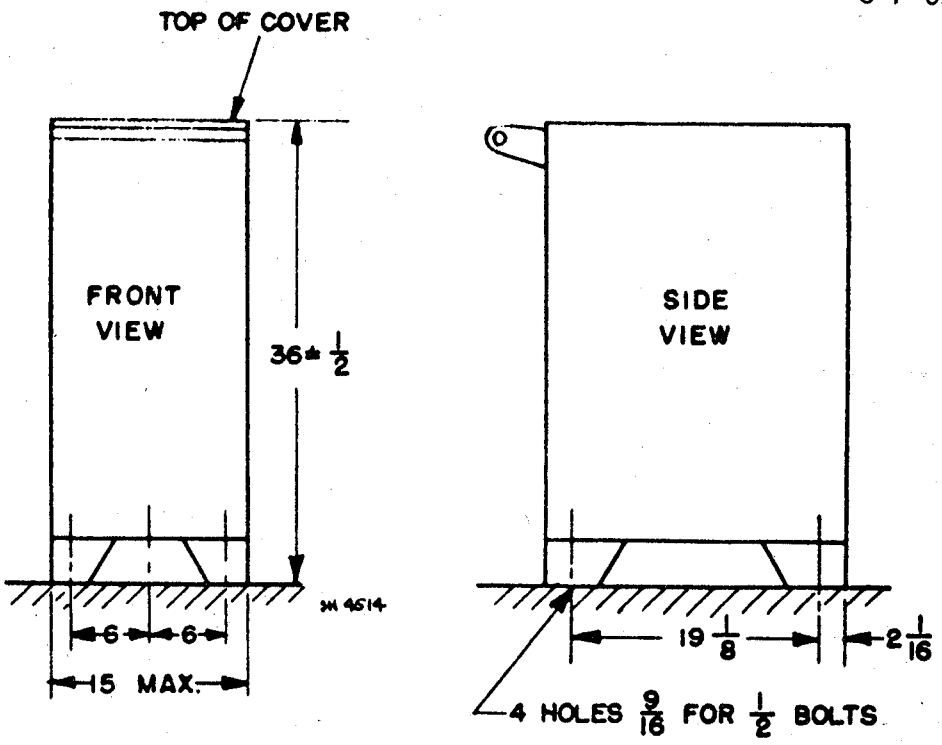


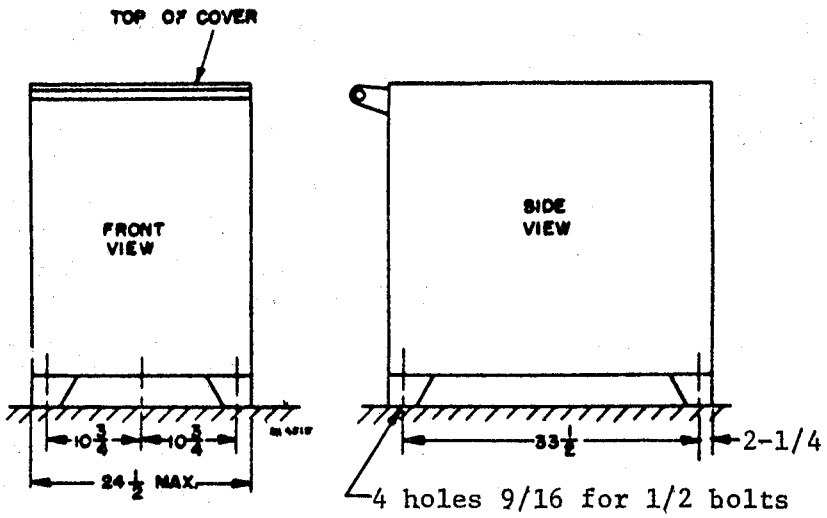
Figure 1 - Dimensions and weight of type I, size 1 fryer.

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NOTES: 1. ALL DIMENSIONS IN INCHES.
2. TOLERANCE: ± 1/16
3. WEIGHT: 180 LBS. MAX.

Figure 2 - Dimensions and weight of type VI fryer



NOTES: 1. ALL DIMENSIONS IN INCHES.
2. TOLERANCE ± 1/16
3. WEIGHT: 260 LBS MAX.

Figure 3 - Dimensions and weight of type II, size 4 and 5 fryer

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S-F-695G

Custodians:

Army - GL
Navy - YD
Air Force - 99

Review activities:

Army - MD
Navy - SA
DSA - GS

User activity:

Army - CE

Preparing activity:

Army - GL

Civil Agency Interest:

GSA-FSS
VA-DMS

Project No. 7310-0644

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